

INDU 211 · Comprehensive Topic Guide (Part 4)

Chapter 5: Material Handling, Distribution & Routing

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering (MIAE)

Built on the teacher's Lecture 5.0 slides, with depth from the course textbook (Hicks, Chapter 5: Material Handling, Distribution, and Routing).

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1. Why Material Handling Matters

Material handling means **using equipment to move materials** inside and between facilities. It is **non-productive (non-value-added)**: moving a part does not change it, yet it costs money, time and space. Chapter 4 already told us that material handling is **30% to 95% of production cost**, which is why layout and handling are designed together.

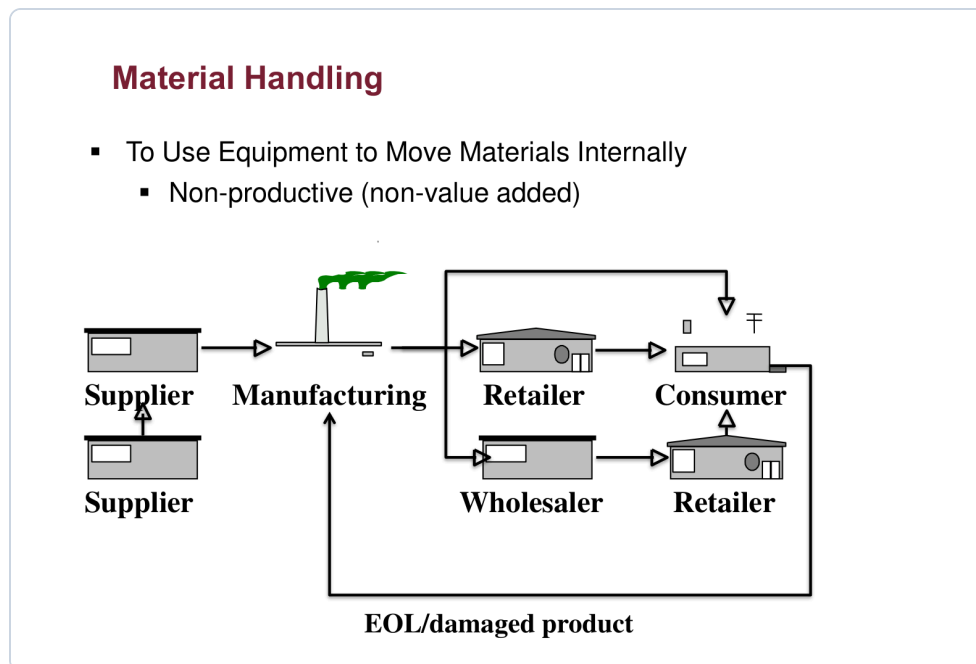


Figure 1: Material handling, Lecture 5.0, slide 3.

Reading the figure: material moves from **suppliers** to **manufacturing**, then to a **wholesaler**, a **retailer** and finally the **consumer**. The arrow returning from the consumer is the reverse flow of **end-of-life or damaged product**. Every arrow is a handling and transport step that adds cost but no value.

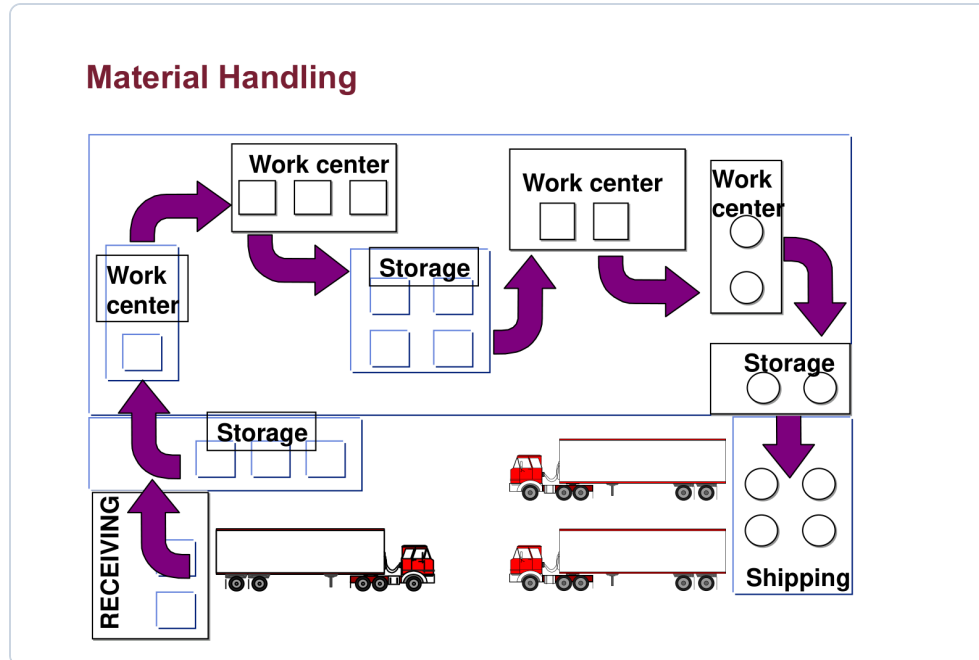


Figure 2: Material handling within a facility, Lecture 5.0, slide 4.

Reading the figure: inside the plant, material arrives at **receiving**, goes to **storage**, travels between **work centres** and intermediate storage, and leaves through **shipping** onto trucks. The purple arrows are the internal moves an industrial engineer tries to shorten, combine or eliminate.

From the textbook (Hicks, §5.1): material movement occurs *any time* a product or part moves from one place to another, and it usually accounts for a large share of the cost of producing the product. Reducing the number and length of moves is one of the cheapest ways to cut cost.

2. Material Handling Equipment

Equipment	What it does	Typical use
Conveyors	Move homogeneous material from one fixed point to another at a constant rate	Assembly lines, parcel sorting
Industrial trucks (forklifts)	Move loads along varying paths , intermittently	Job shops, warehouses
Cranes and hoists	Overhead lifting of heavy loads	Steel mills, heavy assembly
Containers and racks	Store and handle bulk material; better use of space	Pallets, bins, carts
Elevators and lifts	Vertically raise or lower material; fixed location	Multi-floor plants
Automatic guided vehicles (AGV)	Driverless vehicles on predetermined paths	Automated plants, hospitals
AS/RS (automated storage and retrieval)	Storage racks + computer control + crane	High-density warehouses

Material handling Equipment

- Conveyors
 - From one fixed point to another
 - Homogeneous material supplied at a constant rate
- Industrial trucks
 - Varying paths
 - Intermittent load
 - Job shops



Figure 3: Conveyors and industrial trucks, Lecture 5.0, slide 5.

Reading the slide: the roller conveyors (top right) fix the path, which suits steady, identical loads. The forklifts (bottom) can go anywhere with intermittent loads, which suits job shops where every order takes a different route.

The choice of equipment follows the layout: product layouts use conveyors, process layouts use trucks.

Material handling Equipment

- Automatic guided vehicles (AGV)
 - Driverless vehicle
 - Predetermined paths

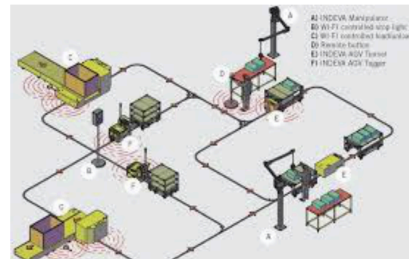


Figure 4: Automatic guided vehicles, Lecture 5.0, slide 8.

Reading the slide: AGVs are driverless carts that follow predetermined paths (wires, tape or laser navigation). The layout sketch (bottom right) shows the guide paths connecting stations: flexible like trucks, but automated like conveyors.

Material handling Equipment

- Automated storage and retrieval system (AS/RS)
 - Storage rack systems
 - Control system (computer)
 - Crane



Figure 5: Automated storage and retrieval system (AS/RS), Lecture 5.0, slide 9.

Reading the photos: tall racks with narrow aisles and a computer-controlled crane that stores and retrieves loads. AS/RS trades a large investment for **space utilisation** (building up, not out) and fast, accurate picking.

3. Principles of Material Handling

Principles of materials handling

- Planning (planned not evolved)
- Systems (material movement cycle)
- Material flow (smooth)
- Simplification (motion economy)
- Gravity (gravity feed bins, roller conveyors)
- Space utilization (high racks with narrow aisle stacking trucks in warehouses)



Figure 6: Principles of materials handling, Lecture 5.0, slide 10 (continued on slide 11).

Principle	Meaning	Example
Planning	Handling is planned, not left to evolve	Handling plan drawn with the layout
Systems	Treat material movement as a cycle from receiving to shipping	Integrated receiving–storage–production–shipping
Material flow	Keep flow smooth	Avoid back-tracking and cross-traffic
Simplification	Motion economy: eliminate, combine, simplify moves	Deliver straight to the point of use
Gravity	Use gravity when possible	Gravity-feed bins, roller conveyors on a slope
Space utilisation	Use the cube, not just the floor	High racks with narrow-aisle stacking trucks
Unit size	Move the largest practical accumulated load	Pallets, containers
Automation	Use powered and automated equipment where it pays	Powered conveyors, automatic pallet stackers
Equipment selection	Choose based on all aspects of the material and move	Weight, shape, frequency, distance
Standardisation	Standard equipment and load sizes	Standard pallets and containers
Adaptability	Equipment that can do a variety of tasks	Variable-speed conveyors
Maintenance	Plan preventive and emergency maintenance	Scheduled conveyor servicing
Safety	Protect the operator	Guarding, safe stacking heights

Exam tip: questions often give an example and ask which principle it illustrates. Gravity-feed bins and roller conveyors → **gravity**; pallets → **unit size**; high racks → **space utilisation**; "planned, not evolved" → **planning**.

4. Routing: The Travelling Salesman Problem (TSP)

Problem: one vehicle leaves a depot (plant A), visits every location exactly once, and returns, so that the **total distance is minimised**. Example: routing a forklift through a warehouse, or one truck supplying all warehouses.

Material routing

- Travelling salesman problem (TSP)
 - Routing a fork truck in a warehouse
- Example 1- the truck must start from the plant (A), visit all warehouses, and return to the plant such that the total travelled distance is minimized.
- One truck can supply all warehouses

To From	A	B	C	D	E	F	G
A	-	14	21	20	6	24	9
B	14	-	10	9	9	10	11
C	21	10	-	1	15	9	21
D	20	9	1	-	14	9	20
E	6	9	15	14	-	19	9
F	24	10	9	9	19	-	21
G	9	11	21	20	9	21	-

Figure 7: Material routing (TSP example 1), Lecture 5.0, slide 12.

Reading the matrix: row = from, column = to. The matrix is symmetric ($A \rightarrow B = B \rightarrow A = 14$), and the diagonal is blank (no trip from a place to itself).

Why a heuristic?

The TSP is **very difficult to solve optimally** when there are many stops: with n stops there are $(n - 1)!/2$ different tours. For 7 locations that is $6!/2 = 360$ tours, which could be enumerated; for 20 it is about 6×10^{16} , which cannot. So we use **heuristics**: fast rules that give a good, feasible route that is not guaranteed optimal.

Nearest Neighbour method (Lecture 5.0, slide 14)

Rule: from the current location, always go to the **closest unvisited** location; when all are visited, return to the start.

Step	From	Closest unvisited	Distance	Running total
1	A	E	6	6
2	E	B (tie with G at 9; take B)	9	15
3	B	D	9	24
4	D	C	1	25
5	C	F	9	34
6	F	G (only one left)	21	55
7	G	back to A	9	64

The optimal tour is **A-G-B-F-C-D-E-A = 60**. Nearest neighbour is $\frac{64-60}{60} = 6.7\%$ worse, and if the tie at E is broken toward G the route becomes 69. **Heuristics can be sensitive to tie-breaking**: the greedy early choices force expensive late legs (F→G = 21).

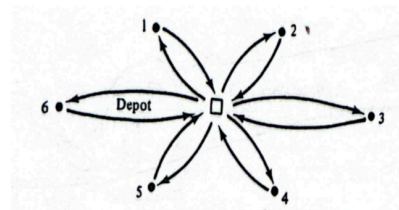
5. Vehicle Routing and the Clark-Wright Savings Method

When one truck cannot carry all the demand, we need **several routes**: the **vehicle routing problem (VRP)**, also called the multiple travelling salesman problem. Each route starts and ends at the depot, and the load on each route must not exceed the **truck capacity**.

Lecture example 2: supply 6 warehouses (B–G) from depot A; truck capacity 25,000 units; demands B 5,000, C 7,000, D 10,000, E 4,000, F 6,000, G 10,000 (total 42,000, so at least $\lceil 42,000/25,000 \rceil = 2$ trucks).

Clark-Wright procedure

- Initial route generation
- Calculate the savings as the result of pairing nodes
- Rank the savings in a descending order
- Start with the first combination to add to the route and continue on the list until no more stop can be added
 - Truck capacity
 - Route feasibility
 - All destinations are visited



$$S_{ij} = C_{oi} + C_{oj} - C_{ij}$$

C_{oi} = distance from depot to stop i
 C_{oj} = distance from depot to stop j
 C_{ij} = distance from stop i to stop j

Figure 8: Clark-Wright procedure, Lecture 5.0, slide 19.

Reading the diagram: the "flower" on the right is the starting solution, with one out-and-back trip from the depot to every stop. Joining stops i and j into one trip removes the two returns to the depot and adds the direct leg $i \rightarrow j$. The distance saved is

$$S_{ij} = C_{0i} + C_{0j} - C_{ij}$$

where C_{0i} is the depot-to- i distance and C_{ij} the distance between the two stops.

The procedure (slide 19)

1. **Initial routes:** one route per stop.
2. **Calculate savings** S_{ij} for every pair.
3. **Rank** the savings in descending order.
4. **Link** pairs in that order, as long as the truck capacity holds, the route stays feasible (a stop can only be joined at the end of a route), and all destinations are eventually visited.

Worked savings: $S_{CD} = 21 + 20 - 1 = 40$ (the largest), $S_{CF} = 21 + 24 - 9 = 36$, $S_{DF} = 20 + 24 - 9 = 35$.

Clark-Wright procedure

Ranked combinations:

<i>Stops</i>	<i>Saving</i>	<i>Stops</i>	<i>Saving</i>	<i>Stops</i>	<i>Saving</i>
1 CD	40	6 BD	25	11 BE	11
2 CF	36	7 BG	12	12 EF	11
3 DF	35	8 CE	12	13 CG	9
4 BF	28	9 DE	12	14 DG	9
5 BC	25	10 FG	12	15 EG	6

Figure 9: Ranked combinations, Lecture 5.0, slide 20.

Building the routes

Rank	Pair	Saving	Decision	Route load
1	C–D	40	Start route Depot–C–D–Depot	17,000
2	C–F	36	Add F at the C end: Depot–F–C–D–Depot	23,000
3	D–F	35	Both already on the same route: skip	—
4	B–F	28	Adding B makes 28,000 > 25,000: reject	—
...	...	...	Continue; build the second route	...
			Second route Depot–E–B–G–Depot	19,000

Final routes: Depot–F–C–D–Depot ($24 + 9 + 1 + 20 = 54$) and Depot–E–B–G–Depot ($6 + 9 + 11 + 9 = 35$), total **89** ($\times 100 = 8,900$ miles).

From the textbook (Hicks, §5.3.2–5.3.3): the same savings logic is used for public-sector routing (garbage collection, school buses, mail). Real VRPs add time windows and driver-hour limits; exact solutions are hard, so heuristics like Clark-Wright and computer software are standard practice (Lecture 5.0, slide 22).

6. Exam Checklist

- [] Material handling is **non-value-added** and 30–95% of production cost.
- [] Match equipment to layout: conveyors (fixed path) vs trucks (varying path) vs AGV vs AS/RS.
- [] Name the principle from an example (gravity, unit size, space utilisation, planning...).
- [] Nearest neighbour: always the closest **unvisited** stop; add the return leg; state the tie-breaking rule.
- [] Savings: $S_{ij} = C_{0i} + C_{0j} - C_{ij}$; rank descending; check **capacity** before every link.
- [] Minimum number of trucks = total demand $\div$ capacity, **rounded up**.